

KIRS, Russian Federation

WMO: 280090

Lat: 59.350N

Lon: 52.250E

Elev: 169

StdP: 99.31

Time Zone: 3.00 (E03)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.7	-29.0	-35.6	0.1	-32.6	-31.9	0.2	-28.9	8.2	-6.0	7.3	-6.6	0.6	160	0.606

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.1	29.1	19.8	27.1	18.9	25.2	17.7	21.0	27.3	19.9	25.3	18.9	23.5	2.4	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.8	13.9	23.4	17.9	13.1	22.5	17.0	12.4	21.5	61.3	27.4	57.8	25.2	54.2	23.7	24.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	6.1	5.2	DB	-36.9	31.7	4.5	1.4	-40.1	32.7	-42.7	33.5	-45.2	34.3	-48.5	35.3	(4)
(5)			WB	-37.0	22.3	4.4	1.2	-40.2	23.2	-42.8	23.9	-45.3	24.6	-48.5	25.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	2.1	-13.3	-12.3	-5.1	2.9	9.8	14.9	17.6	14.5	9.2	2.6	-5.7	-11.2	(6)
(7)		DBStd	12.25	7.90	7.75	5.16	5.03	5.31	4.45	4.03	4.37	4.16	4.35	6.63	8.25	(7)
(8)		HDD10.0	3549	722	624	469	217	73	8	2	8	64	232	472	656	(8)
(9)		HDD18.3	6016	981	858	728	463	268	122	63	137	275	487	722	914	(9)
(10)		CDD10.0	652	0	0	0	5	66	154	237	147	40	3	0	0	(10)
(11)		CDD18.3	77	0	0	0	0	3	17	39	17	1	0	0	0	(11)
(12)		CDH23.3	841	0	0	0	1	58	195	429	152	7	0	0	0	(12)
(13)		CDH26.7	204	0	0	0	0	9	41	110	44	1	0	0	0	(13)
(14)	Wind	WSAvg	2.3	2.3	2.3	2.5	2.4	2.4	2.1	1.7	1.9	2.1	2.5	2.4	2.4	(14)
(15)	Precipitation	PrecAvg	607	37	25	27	35	50	72	80	69	63	47	39		(15)
(16)		PrecMax	878	69	49	67	87	139	205	167	139	118	151	104	71	(16)
(17)		PrecMin	366	7	1	2	9	15	16	18	2	8	7	14	6	(17)
(18)		PrecStd	92	17	9	16	20	27	43	36	33	26	27	20	15	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	0.7	1.7	8.6	21.1	27.7	30.4	31.3	30.9	23.9	17.0	7.3	3.3	(19)
(20)			MCWB	-0.1	0.5	3.5	11.8	16.7	20.7	20.9	20.7	16.0	11.4	6.6	2.9	(20)
(21)		2%	DB	0.0	0.6	5.4	16.7	24.8	27.7	29.5	27.4	20.3	13.3	4.7	1.4	(21)
(22)			MCWB	-0.6	-0.2	1.5	9.6	15.0	19.0	20.1	19.6	14.7	9.4	3.9	0.7	(22)
(23)		5%	DB	-1.4	-0.5	3.3	13.4	22.3	25.7	27.7	24.4	18.0	10.7	2.5	0.4	(23)
(24)			MCWB	-2.0	-1.3	0.6	7.7	13.4	17.6	19.5	18.1	13.6	8.4	1.8	-0.1	(24)
(25)		10%	DB	-3.1	-1.8	1.7	10.8	19.6	23.5	25.8	21.7	16.0	8.9	1.3	-1.1	(25)
(26)			MCWB	-3.7	-2.6	0.0	6.1	12.2	16.6	18.5	17.0	12.7	7.6	0.8	-1.7	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	0.2	1.0	4.1	12.5	18.3	21.9	22.2	21.6	17.2	12.1	6.6	3.1	(27)
(28)			MCDWB	0.6	1.3	8.0	19.6	25.6	28.7	28.8	28.8	21.4	15.5	7.4	3.3	(28)
(29)		2%	WB	-0.6	-0.1	2.3	10.4	16.0	20.2	21.2	20.4	15.6	10.1	3.9	0.9	(29)
(30)			MCDWB	-0.1	0.5	4.4	15.7	22.7	25.7	27.3	26.1	19.3	12.4	4.5	1.2	(30)
(31)		5%	WB	-1.9	-1.2	1.2	8.3	14.5	18.9	20.3	18.9	14.2	8.8	2.0	-0.1	(31)
(32)			MCDWB	-1.4	-0.5	2.7	12.8	20.0	23.8	25.9	23.0	17.1	10.4	2.3	0.3	(32)
(33)		10%	WB	-3.7	-2.6	0.2	6.5	13.1	17.6	19.3	17.6	12.9	7.5	0.9	-1.6	(33)
(34)			MCDWB	-3.2	-1.8	1.6	10.1	18.4	22.2	24.4	21.1	15.6	9.0	1.3	-1.1	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	6.7	8.5	9.2	9.8	11.3	10.8	11.1	9.3	8.1	5.2	4.6	5.6	(35)
(36)			MCDBR	5.5	6.3	10.3	14.5	15.9	14.2	14.1	12.7	11.9	9.1	4.1	3.8	(36)
(37)			MCWBR	5.4	5.8	7.3	8.3	7.4	6.4	5.9	5.5	6.7	5.9	3.9	3.9	(37)
(38)		5% WB	MCDBR	5.4	5.7	7.7	13.5	13.4	12.1	12.7	11.6	10.5	7.9	3.9	3.7	(38)
(39)			MCWBR	5.5	5.6	5.8	8.1	7.0	6.1	5.9	5.6	6.4	5.8	3.8	3.8	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.276	0.308	0.354	0.382	0.374	0.367	0.394	0.396	0.359	0.332	0.291	0.258	(40)	
(41)		taud	2.005	2.036	1.985	2.119	2.372	2.427	2.356	2.329	2.407	2.326	2.121	1.973	(41)	
(42)		Ebn at Noon	478	669	750	810	855	867	829	795	768	661	467	354	(42)	
(43)		Edh at Noon	52	86	123	129	108	104	110	105	82	65	46	36	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.40	1.18	2.36	3.79	4.96	5.35	5.38	3.91	2.26	0.96	0.36	0.22	(44)	
(45)		RadStd	0.06	0.26	0.32	0.51	0.55	0.42	0.55	0.37	0.30	0.12	0.06	0.04	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.73	N/A	N/A	N/A	N/A	N/A	-274	-327	N/A	N/A
(47) Regional (6 neighbors)		+0.60	N/A	N/A	N/A	N/A	N/A	-184	N/A	N/A	N/A

Nomenclature: See separate page